

SRINATH S

AI Engineer | Agentic AI (LangGraph / MCP) | RAG Systems | Machine Learning

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PROFESSIONAL SUMMARY

B.Tech Artificial Intelligence and Data Science graduate specialized in building production-oriented **agentic AI, retrieval-augmented generation (RAG), and NLP systems**. Shipped **ClosePilot**, an enterprise 7-node LangGraph workflow with Model Context Protocol (MCP) tool calling, Claude/NVIDIA LLM reasoning, Human-in-the-Loop gates, and a 14/14 passing Pytest suite. Architected semantic support-ticket intelligence processing **108,819 tickets** with SBERT and FAISS, co-authoring an **IEEE ICIRCA 2026 paper**. Hands-on with Python, async programming, Pydantic v2 schemas, FastAPI, Docker, and AWS EC2 deployment.

WORK EXPERIENCE

Proitbridge — Machine Learning Intern

Sep 2025 – Dec 2025

- Built an end-to-end loan default risk classification pipeline using **CatBoost, pandas, and NumPy** across 100+ consumer features, achieving **90% accuracy and 0.74 ROC-AUC** with hyperparameter optimization.
- Containerized data preprocessing, model inference, and feature engineering modules using **Docker**; deployed on **AWS EC2** for reproducible automated integration testing.
- Integrated **SHAP (Explainable AI)** modules to generate local and global feature attribution reports, communicating risk factor drivers to executive business stakeholders.

SELECTED ENGINEERING PROJECTS

ClosePilot — Multi-Agent AI Sales Copilot (LangGraph, MCP, Claude/NVIDIA, FastAPI, Supabase, Docker)

[\[GitHub Repo\]](#)

- Engineered a stateful **7-node LangGraph multi-agent workflow** (Research, Prioritize, Strategy, Drafter, Approval, Action, Verification) with Pydantic typed state, checkpoint memory, and safe pause/resume.
- Integrated **Model Context Protocol (MCP)** to standardize CRM tool calling with HubSpot; enforced strict **Human-in-the-Loop (HITL) gates** (`interrupt_before=['approval']`) to guarantee zero unauthorized external CRM writes.
- Implemented a zero-hallucination deterministic deal priority scoring formula; added **LangSmith waterfall tracing** and verified end-to-end system reliability with a **14/14 passing Pytest suite**.

Semantic AI Ticket Intelligence Platform (Sentence-BERT, FAISS, scikit-learn, FastAPI, React)

[\[IEEE ICIRCA 2026\]](#)

- Architected a production-grade multi-task support ticket triage system processing **108,819 cleaned enterprise tickets** into 4 structured targets: category classification, routing queue, priority level, and resolution ETA.
- Implemented 384-dimensional **Sentence-BERT (all-MiniLM-L6-v2)** embeddings paired with **FAISS vector indexing** for sub-150ms semantic retrieval of relevant historical ticket resolutions.
- Achieved **81% priority accuracy, 80% action accuracy, and 7.73-hour MAE on ETA regression**; built FastAPI endpoints with SQLite audit logging and co-authored an accepted paper at **IEEE ICIRCA 2026**.

RepoChat — Semantic Repository Intelligence System (LangChain, ChromaDB, Groq LPU, Streamlit)

[\[GitHub Repo\]](#)

- Built an enterprise RAG assistant that semantically parses, chunks, and indexes **1,000+ files per repository** (benchmarked on React and LangChain) into ChromaDB with **<300ms vector retrieval latency**.
- Engineered AST-aware semantic code chunking and grounded generation with **Groq Llama 3.3 70B**; enforced exact line-number citations resulting in **~95% factual accuracy and <2% hallucination rate**.

Redrob Candidate Ranking Engine & YOLOv9 Vision API (BAAI/BGE, Vector Retrieval, PyTorch, YOLOv9)

Hugging Face & Docker

- Redrob Engine**: Precomputed BAAI/BGE dense embeddings for a **100,000-candidate pool**, ranking top 100 profiles against job descriptions in **<8s on CPU-only infrastructure** with deterministic tie-breaking.
- YOLOv9 Vision Service**: Containerized YOLOv9 hazard detection in FastAPI, caching weights to drop latency from **3–5s to 110ms (30x faster)** and reducing Docker image footprint by **55% (4.5GB to 1.9GB)** on AWS EC2.

TECHNICAL SKILLS

Agentic AI & LLMs:

LangGraph (v0.2+), State Machines, Checkpointers, Human-in-the-Loop (HITL), Model Context Protocol (MCP), Anthropic Claude, OpenAI, Groq LPU, Prompt Engineering, Structured Output (Pydantic v2)

Retrieval & Context:	RAG Architectures, Sentence-BERT (SBERT), FAISS, ChromaDB, BAAI/BGE Embeddings, Semantic Search, Context Compaction, Token Budgeting, Line-level Source Attribution
ML & Data Science:	PyTorch, CatBoost, scikit-learn, SHAP (Explainable AI), Multi-Task Classification, Time-to-Resolution Regression, YOLOv9/v8, OpenCV, pandas, NumPy, SQL (PostgreSQL, SQLite, MySQL)
Engineering & Cloud:	Python 3.10+ (async/await, typing, OOP), FastAPI, Uvicorn, Docker, Docker Compose, AWS EC2, AWS S3, LangSmith Tracing, Pytest (14/14 tests), React + Vite, Git, Linux / Bash

EDUCATION & CERTIFICATIONS

B.Tech, Artificial Intelligence & Data Science (GPA: 8.0 / 10)
JCT College of Engineering & Technology, Anna University

2022 – 2026
Coimbatore, India

Verified Credentials: • **Oracle Certified Foundations Associate:** OCI 2023 AI Foundations (ID: 100662362OCI23AIFCA, Valid to 2026) | • **Anthropic:** Introduction to Model Context Protocol (MCP) | • **LangChain Academy:** Introduction to LangChain | • **HackerRank:** Python (Intermediate) & SQL (Basic) | • **IBM:** Python for Data Science & AI, Databases and SQL with Python (Honors)